

Autonomous Theory Construction Laboratory

Lecture 9: Automated Manuscript Generation

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2026-10-26–2026-11-01

Overview

Claims, evidence, figures, and references are assembled from a structured research graph into a conventional manuscript.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$\textit{claim} \rightarrow \textit{evidence} \rightarrow \textit{artifact} \rightarrow \textit{log}$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$\textit{PDF} = \textit{Serialize}(\textit{graph}, \textit{template}, \textit{style})$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$\textit{metadata} = (\textit{authors}, \textit{revisions}, \textit{data}, \textit{software})$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- MethodWriter 4.2 guide
- MAS-54 bibliography profile
- Yun, ch. 9